

Exponentials and logarithms - Set 2

TMUA practice questions by topic. Specification MM5.1-MM5.3. 9 questions.

Each question has one correct answer. No calculator. Answers and worked solutions are in the separate solutions PDF.

1 For which of the following values of x is $3^x < 2^{x+1}$?

I $x = 0$

II $x = 1$

III $x = 2$

A none of them

B I only

C II only

D III only

E I and II only

F I and III only

G II and III only

H I, II and III

2 If $\log_a b = 2$ and $\log_b c = 3$, find $\log_a c$.

A 8

B 6

C $\frac{2}{3}$

D 9

E 5

F $\frac{3}{2}$

3 The graph of $y = 2^x$ is translated 3 units in the positive x -direction. The resulting graph has equation

A $y = \frac{1}{8} \cdot 2^x$

B $y = 2^{x/3}$

C $y = 8 \cdot 2^x$

D $y = 2^x - 3$

E $y = 2^{3x}$

F $y = 2^x + 3$

- 4 Given that $5^x = 2$ and $2^y = 125$, find the value of xy .
- A** 1
B 3
C 5
D 2
E 6
F $\log_5 2 \cdot \log_2 125$
- 5 The number of integers n satisfying $2^n < 1000 < 2^{n+3}$ is
- A** 3
B 4
C 0
D 1
E 10
F 2
- 6 The equation $9^x - k \cdot 3^x + 9 = 0$ has exactly one real solution. Find the value of k .
- A** 3
B -6
C ± 6
D 9
E 6
F 0
- 7 A quantity halves every 5 hours. Which expression gives the number of hours for it to fall to $\frac{1}{10}$ of its initial value?
- A** $5\log_2 5$
B $\frac{\log_{10} 2}{5}$
C $5\log_{10} 2$
D $\frac{\log_{10} 2}{5}$
E 25
F $50\log_{10} 2$

- 8 Which of the following is a counterexample to the claim **'for all positive x and y , $\log_{10}(x+y) \leq \log_{10} x + \log_{10} y$ '**?
- A $x = 10, y = 100$
 - B $x = y = 2$
 - C $x = 5, y = 5$
 - D $x = 4, y = 4$
 - E $x = y = 1$
- 9 Consider the statement: **'for all real x , $2^x > x^2$ '**. Which of the following is a counterexample?
- A $x = 5$
 - B $x = 3$
 - C $x = 0$
 - D $x = 10$
 - E $x = 1$